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Mechanism of Non-Contact ACL Injury

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Abstract

Anterior cruciate ligament (ACL) ruptures significantly impact athletes in terms of return to play and loss of long-term quality of life. Prior to the onset of this research, understanding the mechanism of ACL injury was limited. Thus, the primary focus of this manuscript is to describe our multi-faceted approach to uncovering the mechanism of non-contact ACL injury (NC-ACLI) with the goal of developing preventive strategies. The initial qualitative analysis of ACL injury events revealed most (70%) injuries involve minimal to no contact and occurred during landing or deceleration maneuvers in team sports with a minor perturbation prior to the injury that may disrupt the neuromuscular system leading to poor body dynamics. A series of quantitative videotape studies demonstrated differences in leg and trunk positions at the time of NC-ACLI in comparison to control subjects. Analysis of the faulty dynamics provoking NC-ACLI, especially the flat-footed landing component, supports the theory that an axial compressive force is the critical factor responsible for NC-ACLI. An MRI study demonstrated the NC-ACLI position was associated with a higher tibial slope, and joint contact occurring on the flat, anterior portion of the lateral femoral condyle versus the round, posterior aspect. Both anatomic conditions favor sliding (pivot shift) over rolling in the presence of an axial compressive force. Subsequent cadaveric studies supported axial compressive forces as the primary component of NC-ACLI. Both a strong eccentric quadriceps contraction and knee abduction moments may increase the compressive force at the joint thereby lowering the axial threshold to injury. This manuscript summarizes the NC-ACLI mechanism portion of the 2021 OREF Clinical Research Award.

Keywords

Non-contact ACL injury; axial compression forces; mechanism of injury

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INTRODUCTION

The benefits of physical exercise and sports participation can be tremendous, but the dark side of these activities is the risk of injury, which can prevent return to play, and more importantly, can have life-long consequences, such as osteoarthritis. One of the most common knee injuries, an ACL tear, typically requires surgical reconstruction and extensive rehabilitation for a minimum of six months.[1] The annual cost attributable to the long-term development of osteoarthritis following an ACL injury has been calculated to be \$2.8 billion dollars.[2]

The initial theories proposed to explain the mechanism (causes) of non-contact ACL injury (NC-ACLI) were impingement,[3] vigorous quadriceps contraction,[4] and excessive knee valgus or abduction moments.[5] Impingement of the ACL against the medial border of the intercondylar notch [3] was postulated as an intrinsic cause of NC-ACLI with athletes having intercondylar stenosis being at increased risk.[3] The quadriceps theory attributed NC-ACLI to the anterior vector of a vigorous, eccentric quadriceps contraction generating a force on the proximal tibia, straining and potentially rupturing the ACL.[6] Excessive valgus torque or high abduction moments on the knee, especially in female athletes, was also postulated as a causative factor of NC-ACLI.[5]

One of the limiting components of these early ACL hypotheses was that the conditions surrounding the event could not be exactly replicated in a laboratory setting. As such, it was not possible to directly quantify numerous neurological, mechanical, and conditional elements contributing to ACL injury. Thus, the purpose of this body of work was to elucidate the dynamic factors contributing to a NC-ACLI, using a multi-faceted research approach focused on capturing video data during actual NC-ACLI events, evaluating *in vivo* conditions utilizing MRI, and examining forces about the knee employing cadaver-based methods. The ultimate goal was to provide quantitative insights for effective screening and preventative strategies, and thereby reduce the burden of injury on the individual athlete and society. This manuscript provides an overview of the NC-ACLI portion of the 2021 OREF Clinical Research Award with an emphasis on placing these results into context within our broader knowledge of NC-ACLI.

Qualitative Questionnaire and Preliminary Video Analysis

In an early attempt at understanding the mechanism of ACL injury, qualitative information was summarized based on a questionnaire for athletes post-ACL-injury (n=100) and a descriptive review of videotapes capturing separate ACL injuries (n=27).[7] The data revealed most (70%) ACL injuries transpired with minimal to no contact (both subsequently referred to as NC-ACLI). Injured athletes were in close proximity of opposing players at the time of injury and either possessed the ball or were defending an opponent in possession of the ball.[8, 9] The NC-ACLI occurred immediately following foot-strike with the knee close to full extension, and involved two common athletic maneuvers: a sudden deceleration or a landing motion on a single leg.[10] The injury in our study[7, 8] occurred without an extreme rotational component of the trunk on the leg[7], as had previously been proposed. [11]. This finding was later confirmed by Dai and colleagues [10], who stated “Knee valgus and knee internal rotation motions were not likely significant contributors to the

ACL injury.... (these) rotation angles were likely post injury” This same finding was also confirmed in earlier cadaver[12] and in vivo work.[13]

Combining knowledge of the playing conditions and biomechanics demonstrated that increased demands placed on the neuromuscular system just prior to injury likely disrupted the motor control patterns aimed at protecting the knee during activity, increasing the likelihood of a NC-ACLI.[8] A crucial spark for pre-emptive training programs was the finding that the majority of ACL injuries were non-contact.[7] The lack of significant contact indicated interventions would likely reduce the probability of NC-ACLI. This higher prevalence of NC-ACLI, relative to contact injuries was confirmed by subsequent studies.[14–19] These later studies supported our initial biomechanical findings and reported varying prevalence of NC-ACLI across sports, suggesting sports-specific training could optimize athlete safety. In total, these qualitative studies exposed the multi-faceted components of NC-ACLI.

QUANTITATIVE ANALYSES

Due to the potential for inaccuracies with qualitative analyses, quantitative 2D analysis of a comprehensive videotape database was the natural evolution of the previous studies and this work set the stage for understanding the mechanism of NC-ACLI. In this series of studies, precise measurements of joint angles and trunk position at initial ground contact (IC) and the subsequent 4 still frames was performed using Image J software (NIH, Bethesda, MD).[8, 20, 21] Reproducibility was high with 18 of 20 ICCs greater than 0.95.[8] This constituted an analysis covering 1700 msec from IC (frame rate = 30 frames/sec or 30 HZ). Evaluating the kinematics during injury and comparing it to matched controls provided the first *in vivo* evidence of the whole-body dynamics and potential forces responsible for NC-ACLI.

In the first part of this study videos of athletes, captured during a NC-ACLI event, were analyzed and compared to videos of uninjured athletes performing similar athletic maneuvers.[8] Sagittal plane hip, knee, and ankle angles (Figure 1a) along with coronal plane trunk, hip, and knee angles (Figure 1b) were measured at IC and the four subsequent frames.

In a follow-up study, [21] body kinematics just prior to a NC-ACLI (20 sagittal plane videos) were compared to the body kinematics of control athletes matched for gender, sport, and activity just prior to the analysis frames. The trunk angle, limb angle, and the position of the center of mass relative to the base of support (COM_BOS) were quantified at IC. The COM_BOS was normalized by femur length.

Sagittal Plane

ACL injured athletes initially contacted the ground with the hindfoot or flat-footed, whereas the control subjects landed on the forefoot (Figure 2).[8] The injured athletes (11°) had significantly less ankle plantar flexion than did uninjured control athletes (23°, $P=0.0059$) at IC (Figures 2,3).[8] There was minimal change in the ankle angle (4°) for injured athletes compared to controls (44°) from IC (frame 1) to frame 5.[8] In addition, injured athletes reached the flat-footed position almost 50% sooner than did controls (average 1.6 vs. 2.9

frames).[8] At the knee there was a trend toward less knee flexion in the injured athletes (18°) compared to controls (22°) (Figures 2,3).[8]

Injured athletes demonstrated significantly more hip flexion (50°), than controls (26°) at IC ($p=0.0003$) (Figure 2).[8] The trunk position was significantly more posterior to the base of support in injured athletes.[21] The difference in the average normalized COM_BOS between populations was 0.9 (~38 cm, assuming a femur length of 42cm). The normalized COM_BOS discriminated injured athletes from controls to an accuracy of 80%. No control subject exceeded a value of 1.2 for COM_BOS. When the athlete lands with a relatively flat foot, their body's forward momentum is stopped and the energy is partially converted to angular momentum. The larger COM_BOS in athletes with a NC-ACLI, combined with a relatively fixed tibia and foot position, would naturally cause upper body and thigh rotation to be centered at the knee, fostering posterior translation of the femur relative to the tibia, adding additional strain to the ACL.

Coronal Plane

In the coronal plane (Figure 2) there was no significant difference ($p=0.96$) at IC in knee valgus between the injured (5.5°) versus the controls (5.6°). On average, significant differences in valgus were detected starting in frame 3 (67msec after IC),[8, 20] well beyond the estimated time (17–50 msec) injury occurs after IC.[10, 14, 23–25] At the trunk there was no significant difference at IC in injured versus controls, but female injured athletes did have a higher lateral trunk position compared to male injured athletes ($p=0.02$).[20]

The quantitative video analysis highlights the importance of combined landing leg alignment in NC-ACLI. Knee flexion angles alone do not significantly affect risk of rupture, when compared with uninjured controls. However, a combined landing posture, defined as the “provocative position” (Figure 2) places athletes at greater risk of a NC-ACLI than our defined “safe” position (average limb position of the control group). Although follow-up studies from other researchers are limited in number, they support the hypothesis that low knee flexion angles in combination with increased hip and decreased ankle plantarflexion predispose the athlete to injury.[14, 15] They also support the neutral to slight valgus angulation at IC. This consensus around the provocative position refutes the impingement theory, as injury occurred with slight knee flexion, not hyperextension, as would be expected with impingement. Further, this consensus does not support the notion that valgus is the primary cause of NC-ACLI, although it may secondarily lower the threshold required for NC-ACLI in some athletes.

Axial Compressive Forces are the Primary Component of NC-ACLI

The major breakthrough in understanding the forces responsible for the NC-ACLI occurred when the quantitative biomechanical results were viewed through the lens of the axial compressive force being the critical factor in NC-ACLI. In the safe landing position, the forefoot touches the ground at IC, allowing the calf muscles to absorb the impulsive ground reaction forces (GRF).[23, 26] This is analogous to an airbag in an automobile, which diminish the impact forces before they reach the passenger (Figure 3). In addition, the gastrocnemius contraction likely initiates knee flexion. Therefore, in the safe landing

position the leg acts as an accordion flexing at the ankle and knee to allow the leg muscles to absorb the GRFs. In the provocative (injury) situation, landing flat-footed or nearly flat-footed with minimal knee flexion, renders the calf muscles ineffective at dissipating the GRFs and the impulsive forces bypass the calf muscles and are transmitted directly to the knee.[23] The reduced time span until the injured athlete reaches a flat-footed position also results in increased impulsive forces at the knee by shortening the time for the leg's soft-tissue components to dampen the forces. For a standing athlete weighing 70 kg the GRF is 678N ($9.81\text{m/s}^2 * 70\text{Kg}$). Maximum GRF's recorded by one-leg landing after jumping maneuvers have been estimated to range from 2 to 18 times body weight.[27] Thus, the threshold for ACL injury (2,160N or $3.1 * \text{GRF}$) [28] can easily be exceeded if the calf muscles do not absorb a large portion of the GRFs.[23]

This theory of the triceps surae acting to protect the ACL during a safe landing is supported by previous studies examining single-leg jump-landing tasks using a combined motion capture and musculoskeletal modeling approach.[29] These authors[29] found the soleus reaches 28–32% of the hamstrings posterior force during a jump landing task, providing a protective force for the ACL. Complementary to this, Bakker and colleagues [30] reported that from toe contact to peak ground reaction force, the ankle plantarflexes three times faster than the knee flexes. This rapid rise in the gastrocnemius force seen upon toe contact with the ground, acts to resist dorsiflexion, eccentrically contracting the gastrocnemius, and dissipating the GRFs. In the above-mentioned experiments[29, 30], to preserve participant safety, the exercise studied maintained a safe landing position with the toe touching the floor first, providing time for the ankle to dorsiflex. In the provocative position the athlete lands at or near a flat-footed position and energy dissipation through eccentric contraction is limited.

Tibiofemoral Kinematics

The next unsolved piece of the puzzle was to determine how landing posture, especially hip flexion might affect the tibiofemoral alignment and further contribute to NC-ACLI. In a comprehensive standing MRI study of 25 non-injured athletes the knee was scanned in the sagittal plane. During imaging, the subjects' limbs were placed into positions emulating the safe and then provocative position.[8, 31] The angle between the posterior tibial slope and the femur along with three distances from the tibiofemoral point of contact to: 1. the femoral sulcus point, 2. the posterior tibial point, and 3. the most anterior point of the circular posterior aspect of the condyle were measured for each acquisition.[31]

The tibial slope was significantly ($P < 0.001$) more vertical in the provocative, relative to the safe landing position (Δ tibial slope angle = 13° , $p < 0.001$).[31] This difference was even greater, when measured relative to gravity (Figure 4, Δ tibial slope angle = 30°). In the presence of an axial compressive force at the knee, the increased vertical tibial slope fosters greater posterior sliding of the lateral femoral condyle (LFC) on the lateral tibial plateau, increasing the strain in the ACL and contributing to a potential ACL rupture. This hypothesis is supported by the fact that tibial osteotomies reducing the posterior slope convert an anterior tibial thrust to a posterior tibial thrust with an axial weight-bearing load.[32] These osteotomies have been employed as a treatment for ACL deficiency in canines[23] and

have subsequently become a treatment option for humans with high tibial posterior slopes requiring revision ACL reconstruction.[33]

Analysis of the MRI data also confirmed in the provocative position the point of contact between the LFC and the lateral tibial plateau is significantly ($p<0.001$) more anterior on the LFC, closer to the subluxated position where the bone bruises are located on MRI after NC-ACLI (Figure 5).[31] In the injury position, contact occurs between the flatter anterior portion of the LFC and the convex lateral tibial plateau favoring sliding (pivot shift) over rolling (Figure 7).[31] In the safe landing position, the contact point moves to the rounder posterior aspect of the LFC and normal rolling (knee flexion) is favored over sliding.[31] This propensity for sliding over rolling in the provocative position, similar to 2 ice cubes making contact, would further increase the chances of NC-ACLI. The medial compartment is more stable than the lateral compartment due to the concave medial tibial plateau and the round medial femoral condyle (Figure 6). Therefore, during NC-ACLI sliding occurs on the lateral compartment and rolling on the medial compartment, which leads to an internal rotation moment on the ACL.

Quadriceps Contraction Lowers Axial Threshold to Injury

The cumulative results of our studies support the theory that the impulsive axial, compressive force resulting from a single-leg landing in the provocative position is the main component of NC-ACLI. Yet, this does not exclude a role for excessive quadriceps force and valgus torque, which likely contribute to NC-ACLI by lowering the compressive force threshold for injury. This summary finding is supported by our and others cadaveric work demonstrating excessive joint compressive loads can lead to complete ACL rupture in human knees in conjunction with micro-cracks consistent with ACL bone bruises found on MRI after injury.[34, 35] In one cadaveric study the addition of a quadriceps force reduced the compressive axial forces necessary to produce ACL rupture by 45%.[35]

The posterior trunk position reported in the videotape study[21] indicates an unstable landing in athletes experiencing a NC-ACLI. To stabilize their posture, these athletes need to shift their trunk and thigh forward using the hip flexors and quadriceps muscles. A vigorous quadriceps contraction (Figure 7) lowers the impulse force necessary to tear the ACL by adding a compressive force as well as a smaller secondary anterior shear force on the tibia. Sasaki and colleagues provide support for the validity of our data.[36] The normalized COM_BOS in 60 control female soccer players matches our female controls well (0.72 ± 0.64 vs 0.8 ± 0.5).

Due to the shallow angle of the patellar tendon attachment to the tibial tubercle, it is more plausible the quadriceps contributes to NC-ACLI through its more robust compressive (or superior) vector, which is at least 2-fold higher than the anterior vector at 0° of knee flexion (Figure 7).[23, 35] The bone bruises and impaction fractures [37] seen on MRI after NC-ACLI are also consistent with an axial injury, rather than a shearing force from the anterior pull of the quadriceps.[38]

Knee Abduction Lowers Axial Threshold to Injury

Recent research [26, 39–41] demonstrates the abduction moments at the point of NC-ACLI are less important than was originally proposed.[5] Tibial axial torques induce the highest strain on the ACL regardless of whether an abduction moment is applied or not, based on in vitro [39] and in vivo testing. [13] Our videotape studies reveal no difference in valgus knee position between injured and uninjured athletes at IC.[8] Knee valgus occurs in approximately 50% of athletes with a NC-ACLI after IC, indicating knee abduction moments may play a contributing role, but are not the primary force, which would be present on all NC-ACLI videotapes. The rarity of complete MCL injury at the time of NC-ACLI and the prevalence of medial bone bruising, present in approximately 60% of NC-ACLI at the time of injury, both refute valgus as a primary contributor to NC-ACLI. Rather, the presence of bone bruises on both sides of the tibial plateau support the presence of an axial compressive force as the critical component of NC-ACLI. In addition, Shin and coauthors demonstrated valgus moments increase ACL strain during single leg landing, but are likely insufficient to induce an isolated ACL tear without concomitant damage to the MCL.[42]

However, similar to a quadriceps contraction, valgus alignment likely compounds the effect of an axial compressive load on ACL disruption.[23, 43] By shifting the valgus alignment 2 degrees, the ACL injury compressive force threshold is reduced by the equivalent of 1 body weight.[23] Since females inherently have more knee valgus, this relationship between valgus and compressive load threshold enables females to experience a NC-ACLI at lower GRFs than in male athletes.[40] In addition, any inherent knee valgus, especially in female athletes, may cause tightening of the medial compartment and loosening of the lateral compartment soft tissue allowing the lateral tibial plateau to shift anteriorly with internal rotation increasing the strain on the ACL.[44]

The key role body position plays in NC-ACLI is supported by our prospective NC-ACLI study, which is the largest such study[41] to date with 5758 subjects at 3 large U.S. military centers. Subjects with no history of prior ACL injury are assessed using biomechanical data from a jump landing with the goal of determining risk factors for NC-ACLI. ACL injuries are subsequently tracked for up to 4 years through orthopedic care and medical encounter records. Preliminary (unpublished) results support the concept that hip kinematics at IC of a jump-landing task are associated with subsequent risk of ACL injury. In contrast, this study demonstrates that knee kinematics, including both knee extension and valgus, are not risk factors for NC-ACLI.

Thus, NC-ACLI likely occurs due to a combination of forces straining the ACL. Based on our work, [8, 21, 23, 26, 31] it is clear that the axial compressive load is the driving force behind the majority of NC-ACLI with valgus moments and quadriceps force contributing to the overall load in the ACL. It is also possible that a small percentage of NC-ACLIs occur due to previous minor injuries (micro-tears) which lower the axial threshold injury to failure.[39] In addition, the injury components may vary based on the activity and the athlete. For example, a female athlete with inherent valgus may require less hip flexion or a lower ground reaction force to sustain a NC-ACLI.

Intrinsic versus Extrinsic Factors in NC-ACLI

The combined results of the video and MRI based studies indicate the maneuver being performed at the time of injury has significantly more influence on the likelihood of NC-ACLI than inherent, fixed (non-malleable without surgical correction) risk factors.[26] For example, numerous recent studies demonstrate that an increased anatomic posterior tibial plateau angle (the angle between the tibial plateau and the long axis of the tibia) is a risk factor for NC-ACLI, as this angle is larger by an average of 1.5° [45, 46] in subjects who experience a NC-ACLI, compared to controls. In contrast, based on video analysis, the average difference in dynamic tibial plateau slope between athletes who sustain NC-ACLI compared to controls is approximately 13°, when measured relative to the femur. [26] When the tibial plateau slope is measured relative to gravity, this difference is 30°. Thus, the difference between cohorts for the dynamic slope is 20 times greater ($30 \div 1.5$) than the difference between cohorts based on the inherent slope. Further, a predictive model of multiple inherent risk factors has only produced weak predictability of NC-ACLI.[26, 47] In contrast, assessment of the center of mass in relationship to the base of support on videotape analysis alone can predict with 80% accuracy whether NC-ACLI will occur. [21] This distinction emphasizes the importance of teaching proper leg and body position during landing in ACL prevention programs, while de-emphasizing the effect of fixed, non-malleable factors, such as the inherent tibial plateau slope.[26]

Prevention/Future Research

Our findings have assisted in refining prospective screening tests for ACL injury risk, developing ACL prevention programs, and reducing the incidence of NC-ACLI.[48] Preventive strategies are based on reversing the faulty, straight leg landing position by teaching athletes to land like an accordion on the toes, with the knees flexed, and the chest over the knees.[49] A systematic review of ACL injury prevention programs reported a significant protective effect and reduced ACL injury rates by 53%.[50] Efficacious ACL injury prevention programs included plyometric, strengthening, and agility exercises combined with feedback on proper landing technique.[50]

While our understanding of the mechanism of NC-ACLI has advanced dramatically over the last 25 years, there are still limitations in prior studies and gaps in our current knowledge. First, many of the existing reports rely on small collections of videotapes. With the increased public availability of videos capturing athletic sporting events, access to larger databases for future videotape analyses is possible. Second, ACL injuries disproportionately affect women, yet only a limited number of studies[8, 20, 21] have included women or compared components of landing posture between men and women. Therefore, there is a need for studies evaluating alignment during NC-ACLI in females and studies comparing differences between male and female body alignment during NC-ACLI. A similar case can be made for comparing the mechanism of injury in different sports, especially skiing, which may demonstrate other modes of ACL failure. A third area for improvement is in the use of controls. To date, a limited number of studies [8, 10, 21, 51] compared landing mechanics between controls and athletes with a NC-ACLI. Of these Dai and colleagues [10] present the most interesting path forward. In their study, they were able to capture the same athlete performing the same athletic maneuver on 3 different occasions, with the final

capture, by chance, being acquired during a NC-ACLI. Thus, the strongest evidence, which would serve as an ideal matched control, would be identifying a video of the same athlete prior to suffering a NC-ACLI while performing a similar task to the injury producing task. With numerous training facilities implementing high-quality video captures, future studies would also benefit from transverse plane data to assess whether there is an internal/external rotational component to the NC-ACLI. This is especially important to determine the controversial role of internal rotation, which has been demonstrated to add minimal strain to the ACL during in vivo weight-bearing. [13] In particular, significant amounts of hip rotation can effect sagittal and frontal knee measurements.[52] Lastly, further research is necessary to assess the effect of perturbations prior to NC-ACLI, and whether they can be counteracted through prevention strategies. The fact that preventative programs can reduce the risk of ACL injury by approximately 50% clearly states that the preventative programs, based mainly on the quantitative results of video analyses, are working and need to continue to be applied broadly. We cannot relinquish our quest to further understand the intricacies of NC-ACLI with the goal of further reduction in ACLI through sport and/or athlete specific training.

A key aspect of our NC-ACLI video studies is that they were 2D analyses, either in the sagittal or frontal planes. This likely introduced error due to the slight obliquity in the plane captured relative to the limb segment. A study comparing visual estimates of 2D sagittal plane video analysis to 3D motion capture data[53] reported mean errors of -3° and -16° for hip extension and knee flexion. Interestingly, a more recent study evaluated the Pearson's correlation between 2D angles directly measured from a 2D video still[54] found r-values of 0.51, 0.86 and 0.93 for sagittal plane ankle, knee and hip angles between the 2D and 3D motion capture measures. These strong correlations indicate that much of the previously reported error may be due to the fact that directly measuring 2D angles from video, as was done in our previous studies[8, 21, 22], is more accurate than visually estimating these angles.[53] In terms of the frontal plane, the accuracy of 2D video analysis was better than the sagittal plane[53], but the correlations between the 2D and 3D measures were poor.[54] This indicates the likelihood of more random error, which averages out when taking a mean. As high-quality video from numerous angles during sporting events is becoming rapidly more available, 3D analyses using multiple capture planes offers a promising advancement.[54] Unfortunately, the current 3D video analysis still relies on user interpretation and manipulation of the images. As such, the number of NC-ACLI events that have been analyzed with 3D video analysis is quite limited (no controls) due to excessive analysis times. Thus, work is needed in automating this process both to improve the speed and accuracy of 3D video analysis, and subsequently to determine the accuracy of the 2D compared to 3D video analysis.

Conclusion

In conclusion, our comprehensive ACL injury studies provide strong evidence that axial loading is the primary force responsible for a NC-ACLI. A vigorous eccentric quadriceps contraction and valgus alignment and/or knee abduction moments may increase the strain within the ACL, thereby lowering the axial compressive force threshold for injury. Risk factors for a NC-ACLI are participating in high intensity team sports, running with

possession of the ball or defending the person with the ball, and sustaining a minor perturbation just prior to injury. Minor perturbations may disrupt the normal motor patterns and lead to a faulty leg position at initial contact with the ground. This improper leg position, being an extrinsic factor, has been shown to be malleable and critical to developing ACL prevention strategies. The findings from this research have assisted in refining prospective screening tests for ACL injury risk, enhancing ACL prevention programs, and reducing the incidence of NC-ACLI.

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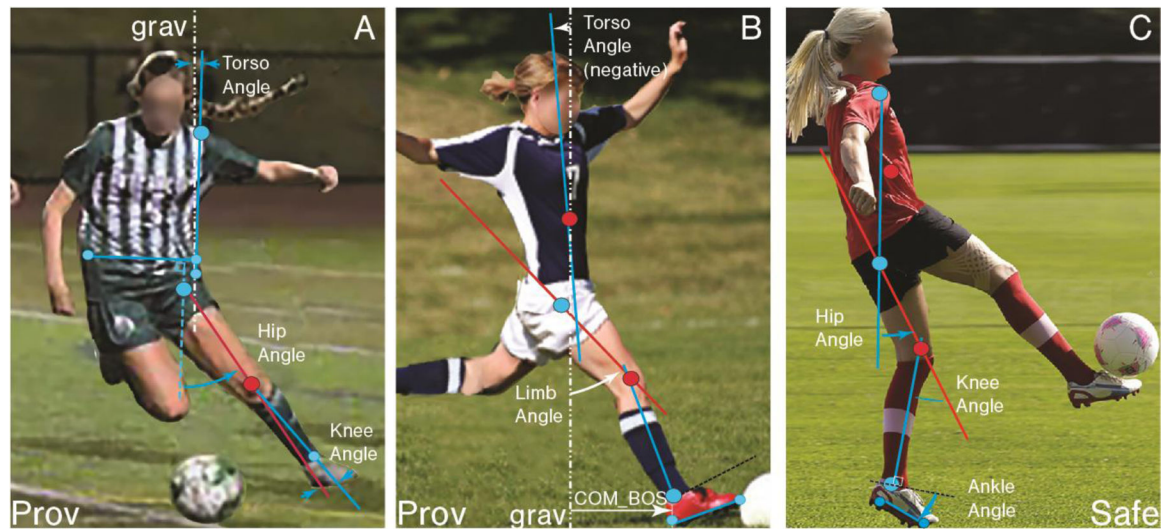


Figure 1. Measures:

A&B) Demonstrate the average provocative position. The dashed-dot line (grav), represents gravity. The average provocative torso angle (B) was $+4^\circ$, but is shown as a negative angle in the figure. **C)** Demonstrates the average control (safe) position. **A)** Coronal plane measures (torso, hip, and knee). [8, 22] **B)** Sagittal plane measures (torso angle, limb angle, and COM_BOS). [21] **C)** Sagittal plane angles (hip, knee, and ankle). [8]

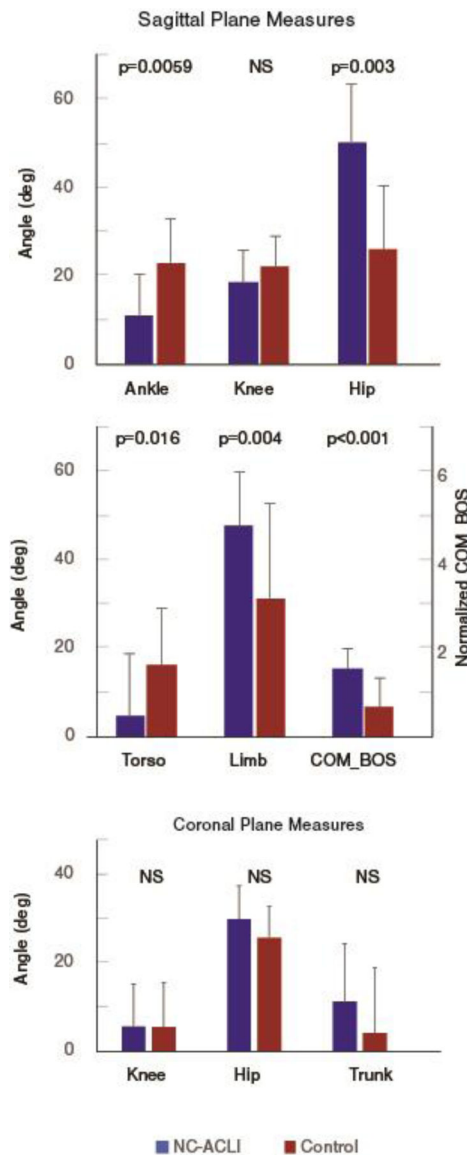


Figure 2. Sagittal and Coronal Plane Landing Biomechanical Results:

Top: Average sagittal plane position at initial contact for NC-ACLI (blue) and control (red) subjects. These measures (Figure 1C) were acquired from videos of 8 female & 3 male during a game situation in which a NC-ACLI occurred, along with videos from 8 female and 4 male controls.[9] **Middle:** These sagittal plane measures (Figure 1B) were acquired from videos of 13 female & 7 male during a game situation in which a NC-ACLI occurred, along with videos from matched controls (13 female and 7 male controls).[19] **Bottom:** Average coronal plane position at initial contact. The first 2 measures (Figure 1A) were acquired from videos of 10 females & 7 males, along with videos from 7 female and 8 male controls.[9] The coronal trunk angle (Figure 1A) was acquired from videos of 10 females, along with videos from 6 female controls.[46]



Figure 3. Dissipation of Ground Reaction Forces:

Calf muscles are analogous to an airbag in a car. Both dampen the impulsive forces prior to reaching either the knee or the passenger.

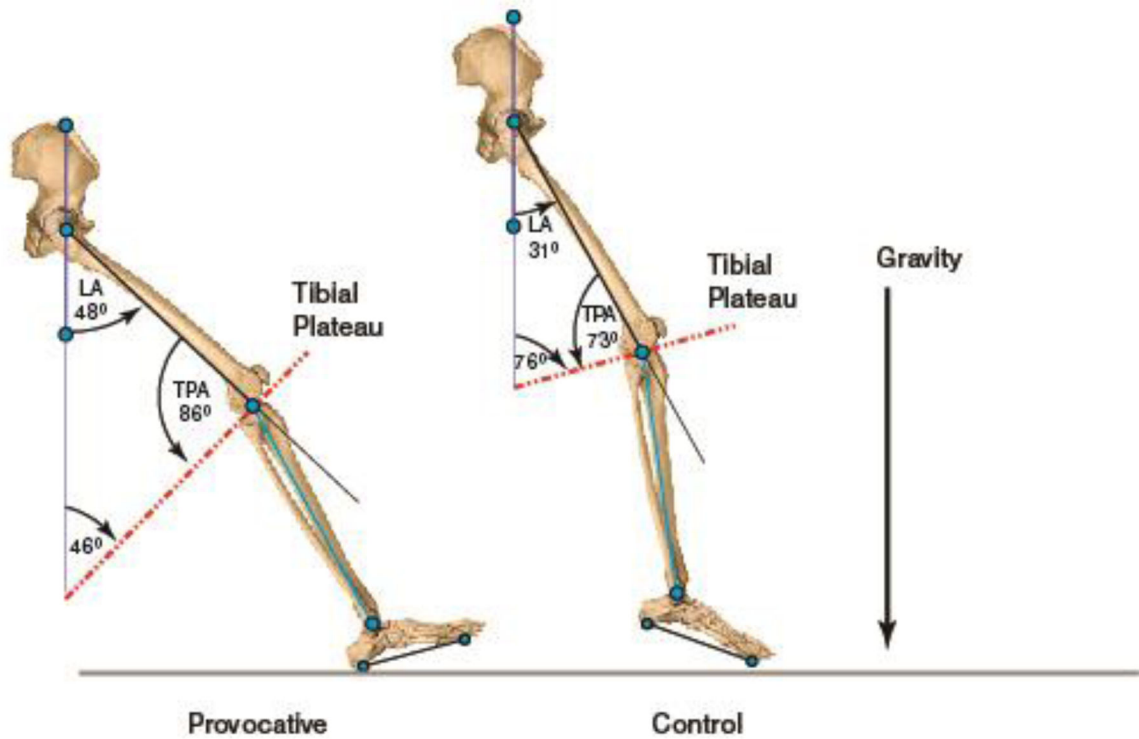


Figure 4. The Tibial Plateau Angle (TPA)

in the provocative vs control positions. On average, the TPA, as measured relative to the femur, was 13° different between cohorts.[31] Referencing the tibial slope to gravity, which accounts for the limb angle (LA), increases this difference to 30°. Note, the knee and ankle are placed in the average provocative and safe positions at initial contact, as reported by Boden et al.[8]

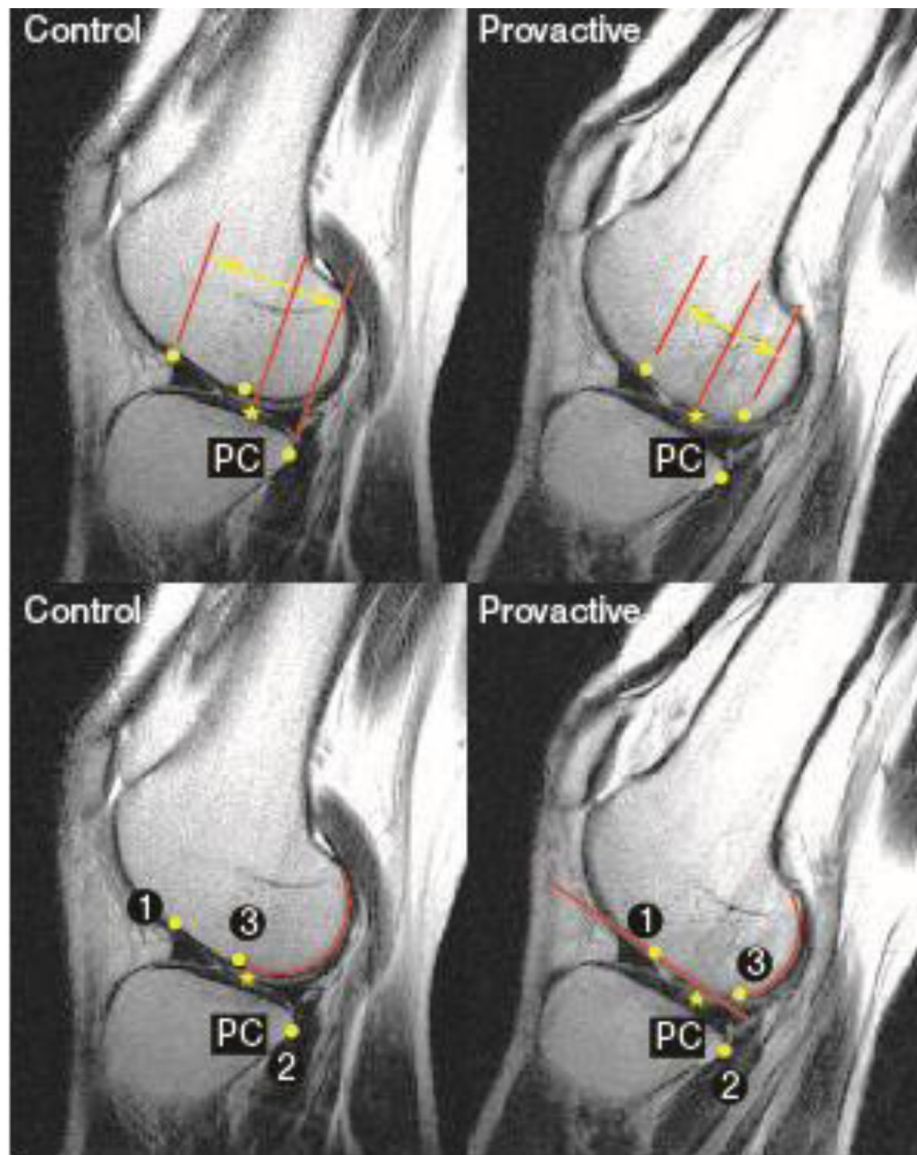


Figure 5. MRI measures of Tibiofemoral Contact Alignment from a Single Subject: Left Column: Safe position. Right Column: Provocative Position. **PC**: Point of Contact, denote as a star. **1**: the femoral sulcus point, **2**: the posterior tibial point, and **3**: the most anterior point of the circular posterior aspect of the condyle. Contact occurs significantly more anterior on the femoral condyle in the provocative position (top row) with the point of contact occurring on the flat portion of the lateral femoral condyle (bottom right), instead of the more circular portion (bottom left).

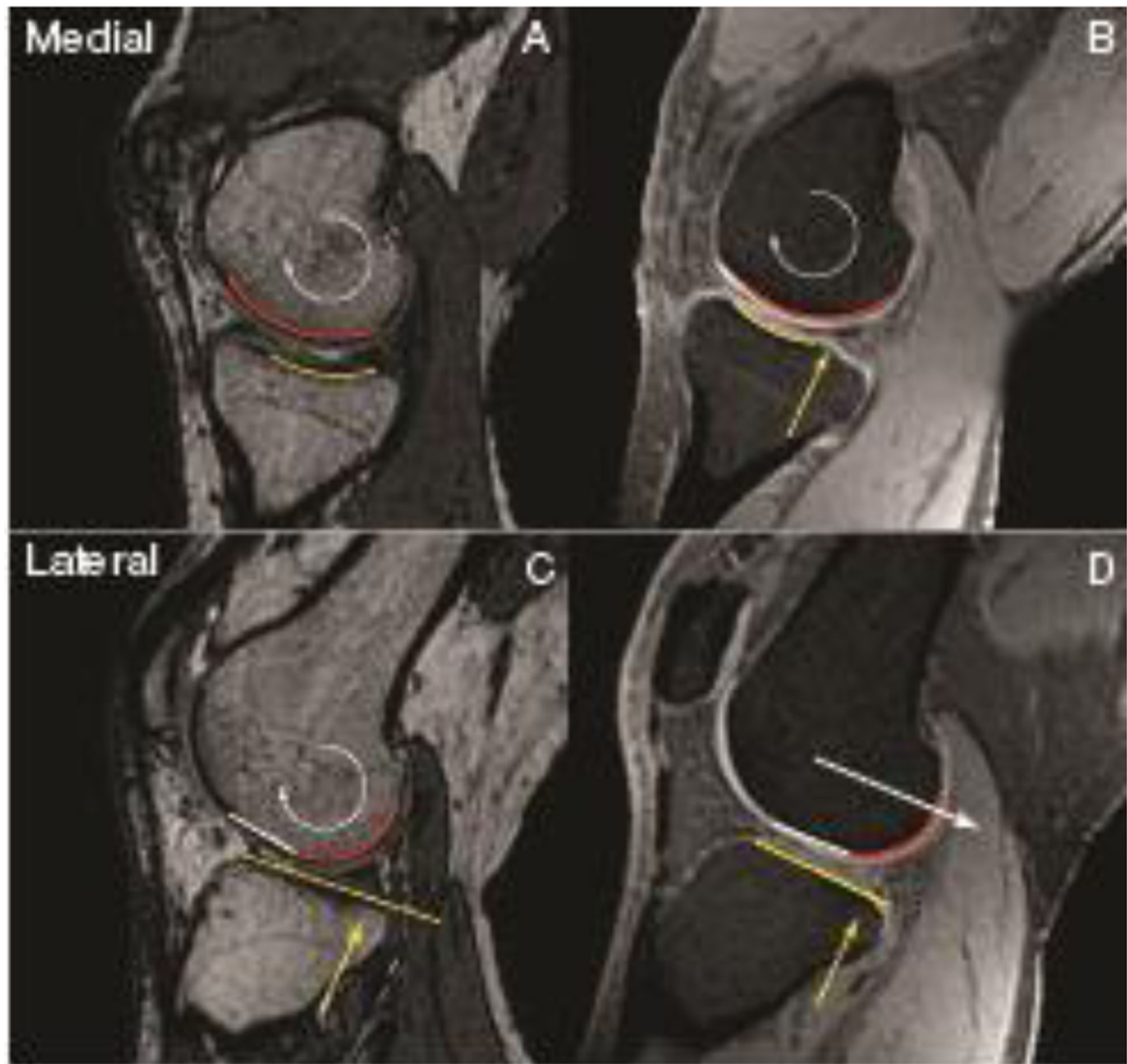


Figure 6. Rolling versus Sliding.

Representation of the point of contact between the tibial plateau and the medial femoral condyle (**top row**) and lateral femoral condyle (**bottom row**) in the safe (**A&C**) and provocative positions (**B&D**). **Medial side:** the round femoral condyle is congruent with the concave (rounded) tibial condyle in both positions, favoring rolling. Even if posterior tibial sliding was forced, the high posterior ridge limits translation. **Lateral side:** with the knee flexed (C), the rounder posterior condyle contacts the convex tibia, favoring femoral rotation/rolling. However, when the knee straightens (D), the flatter bottom edge of the femur contacts the tibia, which does not have a posterior edge to limit femoral translation. In this position (D) sliding is favored over rolling.

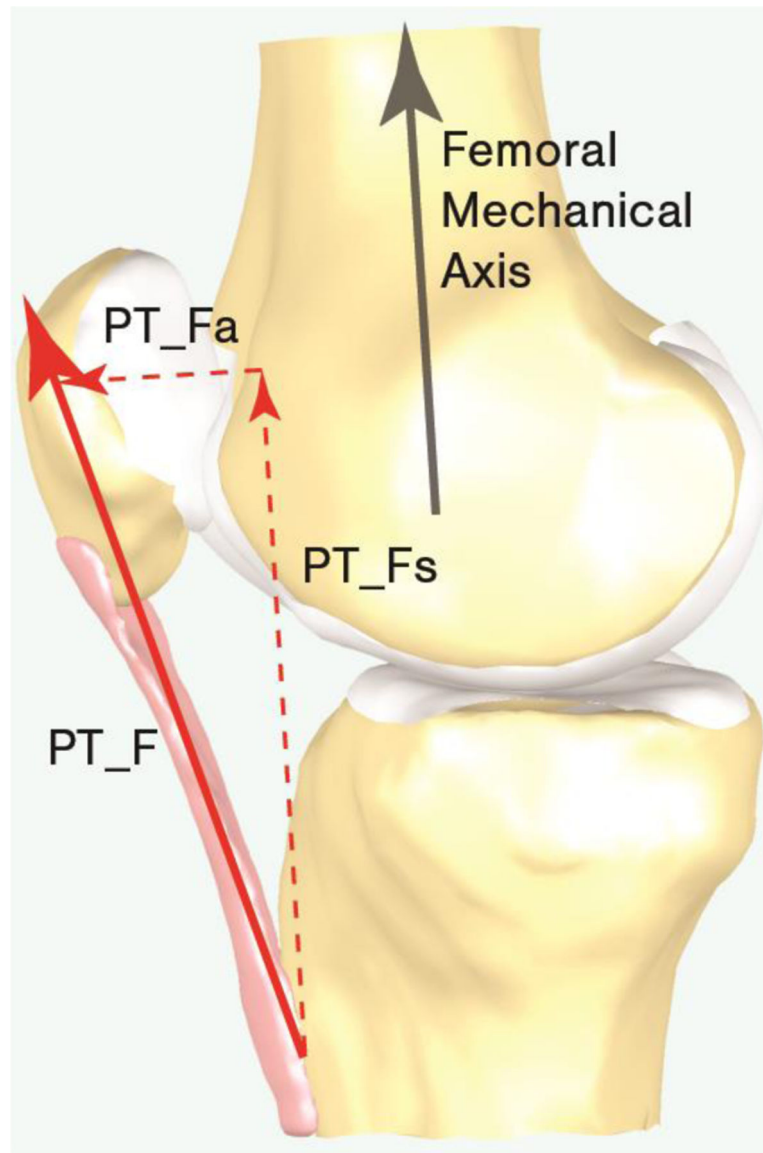


Figure 7. Quadriceps/Patellar Tendon Force on the Tibia:

. The quadriceps exert a force on the tibia through the patellar tendon. As the angle of the patellar tendon relative to the mechanical axis of the femur is low in terminal extension, the primary component of the patellar tendon force (PT_F) is directed superiorly (PT_Fs), which contributes to the overall axial compression force. The component of the force directed anteriorly (PT_Fa) provides minimal sheering force, in comparison.